Machine Learning: glossary

Every word in the course, with the session where it first appears. This list is generated from the "Words for today" cards, so a word cannot be taught without reaching this handout.

Word	What it means	Think of it like this	Session
accuracy	The share of predictions that are right.	The shots that go in, out of all the shots.	ML16
actual and predicted	Rows show what really happened; columns show what the model said.	The real result against the guess.	ML17
AI use log	A record of any assistance, advice taken or rejected, and tests run.	A signed note of who helped.	ML21
argument	The actual value you give the function when you use it.	What you write in the blank.	ML06
assert	Stops with an error if a statement is false.	A friend checking that an answer is right.	ML06
audience	The people listening and what they already know.	The parents in the room.	ML23
axis label	Words that name what an axis measures, with its unit.	The heading on a column in a register.	ML11
backup	A screenshot or saved file you can show if the laptop fails.	A spare key.	ML23
bar chart	Bars compare a number across categories.	Stacks of coins, one stack per item.	ML11
baseline	A simple guess a model must beat, such as the training mean.	Guessing that every home costs the average rent.	ML13
baseline error	The average miss of the plain guess on validation.	How wrong the average guess is.	ML20
baseline model	A model that guesses the training mean every time.	The average guess.	ML15
Boolean	A value that is either True or False.	A light switch that is on or off.	ML04
box plot	A box shows the middle half of the values, with a line at the median.	A summary of a team's scores in one picture.	ML11
candidate model	A model you are considering using.	A player trying out for the team.	ML21
certificate	A programme certificate for finishing the camp: most classes, a build log and a finished project.	A badge for finishing a trail.	ML24
check	A test with a known right answer.	Weighing a bag of sugar on a trusted scale.	ML06
claim	A statement that needs a visible result.	I say the model predicts fares to within about UGX 3000.	ML23

Word	What it means	Think of it like this	Session
class label	The name of a category; its code number is only an identifier.	The species name on a seed packet.	ML16
classification	Predicting a category instead of a number.	Deciding whether a fruit is a mango or an orange.	ML16
cleaning log	A list of every change made and why.	The marking notes beside a corrected exercise.	ML10
code cell	A box where you type Python and press Shift+Enter to run it.	A calculator that sits inside your page.	ML01
coding assistant	A tool that suggests code or fixes; its advice must be checked.	A friend who sounds sure and is sometimes wrong.	ML21
comment	A line starting with # that Python ignores and people read.	A note written in the margin.	ML03
comparison	A question about two values that answers True or False.	Asking whether your money is at least the ticket price.	ML04
confusion matrix	A table that counts actual categories against predicted categories.	A tally sheet of right and wrong guesses, kept in boxes.	ML17
convert	Change a value from one type to another, such as text to a number.	Changing shillings into dollars before adding.	ML03
CSV file	A plain text file where each line is a row and commas split the columns.	A register copied as text.	ML07
data dictionary	A list of each column with its meaning and unit.	The key at the bottom of a map.	ML12
DataFrame	The pandas name for a table inside Python.	A spreadsheet that lives in the notebook.	ML07
decision tree	A chain of yes or no questions about the features that ends in a category.	A flow chart that sorts the flowers.	ML16
demonstration	Showing your own work running or explaining its steps.	Showing the dish you cooked.	ML23
depth	How many questions in a row the tree may ask.	How many forks a path may take.	ML16
dictionary	Pairs of a name and a value inside curly brackets.	A phone book of names and numbers.	ML05
disclosure	Saying where help or AI was used.	Thanking those who helped.	ML23
duplicate	A row repeated exactly. It may be an accident or a real second sale.	The same receipt copied twice.	ML10
elif	Asks a second question when the first was False.	If it is raining take an umbrella, otherwise if it is cloudy take a hat.	ML04
else	Runs when the if question was False.	Otherwise, leave the umbrella.	ML04
error	The gap between a prediction and the real value.	How far off your estimate was.	ML14

Word	What it means	Think of it like this	Session
error message	Python telling you what it could not do and where.	A friend pointing to the exact line that went wrong.	ML03
fair comparison	Same rows, same measure, same split for every model.	Every runner on the same track.	ML21
feature	An input known before you predict.	The clues in a puzzle.	ML13
feedback	A strength and a next step from the instructor.	A coach's tip after training.	ML24
file	A named container for your work on the laptop.	A paper file in a folder.	ML01
filter	Keep only the rows where a comparison is True.	Picking out the sales of 5 units or more.	ML08
finding	A claim a chart supports, with a number.	Book earns the most money: UGX 297000.	ML11
fit	Let the computer find the best slope and intercept from examples.	Practising until the throws land close.	ML14
float	A number with a decimal part.	A price of 2500.5 shillings.	ML02
frozen choice	A decision fixed before the held-back set is used.	Sealing your prediction in an envelope before the match.	ML18
frozen model	A model whose inputs and settings are fixed before the held-back run.	A recipe you stop changing before you serve it.	ML22
function	A named set of instructions you can use again and again.	A recipe card you reuse.	ML06
group	Gather rows with the same label before calculating.	Sorting the sales into three piles by item.	ML08
handover	Everything another person needs: data origin, code, instructions and limits.	Passing on the keys with directions.	ML22
header	The name written at the top of each column.	The title above a column in an exercise book.	ML07
held-back run	The one honest score, run once after all choices are fixed.	Opening the sealed envelope.	ML22
held-back set	The last rows, kept apart and used once after every choice is made.	A sealed envelope you open at the end.	ML17
held-out records	The rows kept back and never used to fit.	A pile of cards kept in a drawer until the end.	ML15
histogram	Bars show how many values fall in each range.	Sorting pupils into groups by height.	ML11
if	Runs the indented lines below it only when its question is True.	If it rains, take an umbrella.	ML04
indentation	Four spaces at the start of a line that show which instructions belong together.	Writing a sub-point under a heading.	ML04

Word	What it means	Think of it like this	Session
index	The position of an item in a list, counting from 0.	The house number of a plot.	ML05
index=False	Tells to_csv not to write the extra row-number column.	Not copying the page numbers.	ML10
input	A measurement known before you predict the target.	The clues you are given.	ML12
inputs	The columns known before the prediction.	The clues you are given.	ML19
integer	A whole number with no decimal part.	Counting 3 books.	ML02
intercept	The prediction when the input is zero.	The starting fee before any distance is added.	ML14
item	One value taken from a list during a loop.	The pupil currently getting a book.	ML05
key	The name used to look up a value in a dictionary.	The name you look up in a phone book.	ML05
leakage	Using information that would not be known at prediction time.	Reading the last page of a mystery before you start.	ML13
leakage column	A column only known after the answer, so it must not be an input.	A tip that is only known after the ride.	ML19
library and import	A library is code other people wrote. import loads it into your notebook.	Borrowing a toolbox from the workshop.	ML07
limit	A fair statement of what the result cannot show.	The edge of what you know.	ML22
limitation	What the data or method cannot tell you.	The edge of the map.	ML12
limits	What the data and the model cannot tell us.	The edges of the map.	ML19
list	Several values kept in order inside square brackets.	A queue of customers.	ML05
live change	One small edit to your own code, made while people watch.	Adjusting a recipe while friends look on.	ML24
loop	A for statement that repeats indented lines for every item.	Giving each pupil in a row a book.	ML05
majority baseline	A model that always names the most common category.	Always answering the most popular option.	ML16
mean	The total shared out equally: add the values and divide by how many.	Sharing a bag of mandazi equally.	ML09
mean absolute error	The average size of the misses, in the target's own unit.	The average distance of your darts from the bullseye.	ML15
median	The middle value once the values are in order.	The pupil standing in the middle of a line sorted by height.	ML09

Word	What it means	Think of it like this	Session
missing value	A cell with no recorded value. Its true value is unknown.	A blank in the price column.	ML10
new column	A column calculated from other columns, added to the table.	Multiplying units by price on every line.	ML08
next step	One learning goal for after the camp.	The next chapter you plan to read.	ML24
notebook	A page made of text cells that explain and code cells that run.	An exercise book where the answers can run themselves.	ML01
operator	A symbol that tells Python which calculation to do.	The plus and minus keys on a calculator.	ML03
outlier	A value far from the rest.	One huge order in a week of small sales.	ML09
output	What Python shows under a cell after it runs.	The answer a calculator shows after you press equals.	ML01
overfitting	Learning the quirks of the training examples so well that new examples are predicted worse.	Learning one route by heart and getting lost on a new road.	ML17
parameter	The name inside the function for the value it receives.	The blank on a form.	ML06
parentheses	Brackets that make Python calculate the inside first.	Doing the part in brackets first in maths.	ML03
peeking	Looking at the held-back set while you choose settings, which spoils it.	Reading the last page of a mystery first.	ML17
percentile	The value below which a given share of the data falls.	The mark that 75 out of 100 pupils did not beat.	ML09
predict	Use the fitted line on a new input.	Reading the answer off the line.	ML14
prediction	A number or label a computer guesses for something it has not seen yet.	A fundi estimating a job before measuring it.	ML01
print	A ready-made instruction that shows a value on the screen.	Reading a number out loud.	ML01
project folder	The folder of your own project files, your build log and your feedback.	Your best work in one box.	ML24
project plan	A short page that fixes the question, data, target, inputs, baseline, held-back set and limits before any building.	A drawing of the house before laying bricks.	ML19
proportion	A part divided by the whole; times 100 it becomes a percentage.	12 pupils out of 20 passed.	ML09
protected rows	Rows no choice may use until the end.	A sealed envelope you open once.	ML20
random_state	A number that makes the shuffle repeatable.	Giving everyone the same shuffled deck.	ML15

Word	What it means	Think of it like this	Session
range	The largest value minus the smallest.	The gap between the tallest and the shortest.	ML09
raw data	The file as you received it. Keep it unchanged.	The original receipt book.	ML10
regression	Predicting a number.	Estimating a price.	ML14
reproducible	Someone can rerun your notebook from the top and get the same results.	A recipe anyone can follow.	ML12
restart and run all	Restart the kernel and run every cell from the top.	Starting the game from the first level.	ML22
return	Sends a result back so the program can keep it.	Handing over the finished item.	ML06
revenue	Money received from sales: units times price. It is not profit.	The takings before paying for stock.	ML08
row and column	A row is one record across the table. A column is one kind of fact down the table.	One line of a register and one heading of the register.	ML07
scatter plot	Each row is a dot; its place shows two values.	Dots on a map of distance and fare.	ML20
select a column	Name the column in square brackets to look only at it.	Reading only the price line of the register.	ML08
shape	The number of rows and columns, written as a pair.	Counting how many lines and headings a register has.	ML07
show-and-tell	A short, friendly turn to show the group something you built.	Showing a friend your drawing.	ML12
showcase	A short event where you present your project to parents and guardians.	An open day at a workshop.	ML24
slope	How much the prediction changes when the input rises by one unit.	Extra rent for every extra square metre.	ML14
split sizes	How many rows go to training, validation and test.	How the class is divided for the three jobs.	ML18
stratify	Keep the share of each category the same in every part.	Making sure every group has some of each village.	ML18
string	Text inside quotation marks.	A name written on a label.	ML02
sum	All the values added together.	Adding up the day's takings.	ML08
target	The value you want to predict, such as monthly rent.	The number you are guessing.	ML12
test data	Examples held back to check the model on things it has not seen.	A box of examples kept aside to try the model on.	ML13
test_size	The share of rows kept back for testing.	One quarter of the cards go in the drawer.	ML15

Word	What it means	Think of it like this	Session
three-way split	Training, validation and held-back parts.	Practise, rehearse, then perform.	ML20
title	One line saying what the chart shows.	The heading of a notice.	ML11
to_csv	Saves a table into a CSV file.	Copying the register to a new page.	ML10
traceback	The list of lines Python followed before the error.	A trail of footprints that ends at the problem.	ML03
train and test split	Divide the rows into homes to learn from and homes to try the model on.	One pile to practise on and one pile kept for trying it out.	ML15
training data	The examples the model learns from.	The worked examples in a recipe book.	ML13
training only	Fit anything only on the training part.	Practising only with your own pile of examples.	ML20
type	The kind of value: integer, float or string.	Knowing whether a box holds pens or juice.	ML02
validation error	The average miss on validation rows.	The score from a trial run.	ML21
validation records	Held-out rows used to choose between models.	A trial run used to decide which plan to follow.	ML17
validation score	A score on held-out rows used to choose between models.	A trial run.	ML18
value	One piece of information, such as 2000 or Juice.	One entry in a receipt.	ML02
variable	A name that points to a value.	A label on a box of stock.	ML02